50Days-coding-challenge

Day-28[Python]

Data Structures Coding Challenge

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1.STRING Challenge

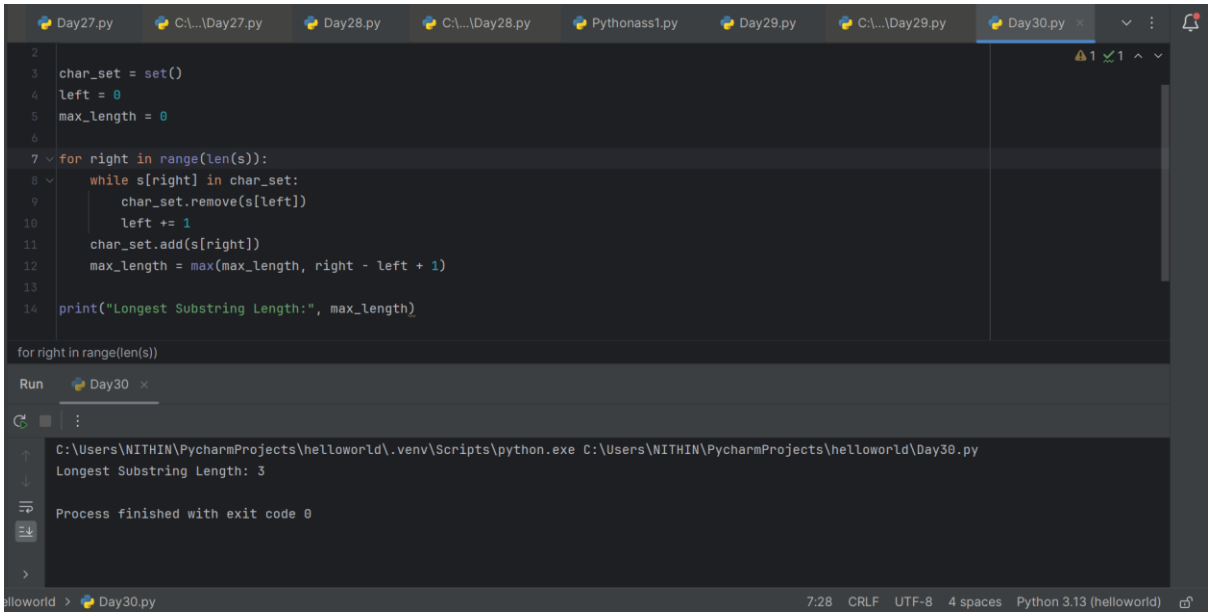
Problem: Longest Substring Without Repeating Characters

Description:

Given a string, find the length of the longest substring without repeating characters.

Input:

"abcabcbb"



2.LIST Challenge

Problem: Remove Duplicates While Preserving Order

Description:

Given a list of integers, remove duplicates while maintaining the original order.

Input:

[1, 2, 2, 3, 4, 3, 5]

```python
max_length = 0

for right in range(len(s)):
    while s[right] in char_set:
        char_set.remove(s[left])
        left += 1
    char_set.add(s[right])
    max_length = max(max_length, right - left + 1)

print("Longest Substring Length:", max_length)

lst = [1, 2, 2, 3, 4, 3, 5]
```

```
C:\Users\NITHIN\PycharmProjects\helloworld\.venv\Scripts\python.exe C:\Users\NITHIN\PycharmProjects\helloworld\Day30.py
Longest Substring Length: 3
List after removing duplicates: [1, 2, 3, 4, 5]

Process finished with exit code 0
```

3.TUPLE Challenge

Problem: Find Maximum and Minimum

Description:

Given a tuple of numbers, find the maximum and minimum values.

Input:

(10, 5, 20, 8)

```python
lst = [1, 2, 2, 3, 4, 3, 5]

result = []
for num in lst:
    if num not in result:
        result.append(num)

print("List after removing duplicates:", result)

tup = (10, 5, 20, 8)

print("Maximum:", max(tup))
print("Minimum:", min(tup))
```

```
Longest Substring Length: 3
List after removing duplicates: [1, 2, 3, 4, 5]
Maximum: 20
Minimum: 5

Process finished with exit code 0
```

4.DICTIONARY Challenge

Problem: Character Frequency Counter

Description:

Given a string, count the frequency of each character and store it in a dictionary.

Input: "programming"

```python
print("Minimum:", min(tup))

s = "programming"

freq = {}

for char in s:
    if char in freq:
        freq[char] += 1
    else:
        freq[char] = 1

print("Character Frequency:", freq)
```

```
List after removing duplicates: [1, 2, 3, 4, 5]
Maximum: 20
Minimum: 5
Character Frequency: {'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}

Process finished with exit code 0
```

Challenge (Intermediate) Problem: Group Words by Length

Description:

Given a list of words, group them based on their length using a dictionary.

Input:

["cat", "dog", "elephant", "bat", "tiger"]

```python
print("Character Frequency:", freq)

words = ["cat", "dog", "elephant", "bat", "tiger"]

grouped = {}

for word in words:
    length = len(word)
    if length not in grouped:
        grouped[length] = []
    grouped[length].append(word)

print("Grouped Words:", grouped)
```

```
Maximum: 20
Minimum: 5
Character Frequency: {'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
Grouped Words: {3: ['cat', 'dog', 'bat'], 8: ['elephant'], 5: ['tiger']}

Process finished with exit code 0
```